Retinoids 126

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Key features

- Retinoids are structural and functional analogues of vitamin A that exert multiple effects on cellular differentiation and proliferation, the immune system, and embryonic development
- Retinoic acid receptors (RARs) and retinoid X receptors (RXRs) are ligand-dependent transcription factors belonging to the steroid–thyroid hormone nuclear receptor superfamily
- *Topical* retinoids are used to treat a number of conditions including acne, photoaging, and psoriasis; the limiting factor is dose-related skin irritation, and teratogenicity risk is very low to non-existent
- *Oral* retinoids are the most effective drugs available for severe acne (isotretinoin) and many disorders of cornification (acitretin and isotretinoin). They are also helpful in the management of psoriasis, especially pustular and erythrodermic forms (acitretin > isotretinoin), pityriasis rubra pilaris (acitretin and isotretinoin), cutaneous T cell lymphoma (bexarotene), and chronic hand eczema (alitretinoin)
- Systemic retinoid therapy requires careful patient selection and monitoring due to associated teratogenicity and other potential toxicities; laboratory abnormalities (e.g. hypertriglyceridemia, elevated transaminases); ocular, musculoskeletal, and (primarily for bexarotene) endocrinological or hematologic adverse effects

INTRODUCTION

Vitamin A (retinol) and related compounds with either structural (retinol derivative) or functional (vitamin A activity) similarities are known as retinoids (Fig. 126.1). The importance of retinoids in cutaneous biology was first appreciated early in the twentieth century by Wolbach, who observed that vitamin A-deficient animals manifested altered keratinization, such as epidermal hyperkeratosis and squamous metaplasia of mucous membranes. The anti-keratinizing properties of vitamin A led von Stuettgen and Bollag to administer topical and systemic retinoids to treat disorders of cornification. In initial clinical trials with oral vitamin A, tolerable doses were ineffective but higher, potentially effective doses were too toxic. Because of this narrow therapeutic window, a program was launched to engineer synthetic retinoids with higher therapeutic activity and lower toxicity. Parallel to efforts with systemic agents, a growing interest in topical retinoids emerged.

All-*trans*-retinoic acid (a*t*-RA; tretinoin), a naturally occurring metabolite of retinol (see Fig. 126.1), was the first retinoid to be synthesized. Orally, it did not have significant advantages over vitamin A, and, as a result, the primary dermatologic focus became its topical use, especially for acne vulgaris and photoaging.

Initially synthesized in 1955, 13-*cis*-retinoic acid (13-*cis*-RA; isotretinoin) was first studied clinically almost two decades later. Its use was delayed by concerns regarding its teratogenicity, which were amplified because of the thalidomide tragedy. Results of isotretinoin treatment of psoriasis were equivocal, and etretinate was subsequently found to be more effective. In the mid to late 1970s, Peck and colleagues described the effectiveness of oral isotretinoin in the treatment of lamellar ichthyosis and other disorders of cornification as well as its ability to produce complete responses with prolonged remissions in patients with treatment-resistant nodulocystic acne[1,2]. Topical isotretinoin has also been formulated and has efficacy similar to that of topical tretinoin for acne. In 1999, the US Food and Drug Administration (FDA) approved 9-*cis*-retinoic acid (9-*cis*-RA; alitretinoin) 0.1% gel for the topical treatment of cutaneous Kaposi sarcoma. To date, oral alitretinoin has been approved outside of the US to treat severe, recalcitrant hand eczema.

In 1972, Bollag developed two aromatic retinoids: etretinate and its free acid metabolite, acitretin. Both had a therapeutic index ten times more favorable than that of tretinoin in the chemically induced rodent papilloma test model. Etretinate and acitretin represent second-generation retinoids, and their development constituted a breakthrough in the systemic treatment of psoriasis and disorders of cornification.

A major advance in understanding the molecular pharmacologic features and mechanisms of action of retinoids came with the discovery of nuclear receptors known as retinoic acid receptors (RARs) and retinoid X receptors (RXRs). This allowed the development of new retinoids that target specific retinoid receptors. FDA-approved third- and fourth-generation retinoids include topical adapalene (acne), topical tazarotene (psoriasis and acne), topical trifarotene (acne), and oral as well as topical bexarotene (cutaneous T cell lymphoma [CTCL]). Lastly, topical precursors of retinoic acid such as retinyl esters, retinol and retinaldehyde are included in cosmeceutical products.

MECHANISM OF ACTION

Retinoids are involved in the regulation of diverse biologic functions (Table 126.1). They can: (1) affect cellular growth, differentiation and morphogenesis; (2) inhibit tumor promotion and malignant cell growth; (3) exert immunomodulatory actions; and (4) alter cellular cohesiveness[3].

Vitamin A Metabolism

Vitamin A (retinol) must be acquired through the diet as retinyl esters and provitamin A carotenoids; of the latter, β-carotene is particularly efficient in its ability to be converted to vitamin A. Within the intestinal lumen, retinyl esters are hydrolyzed to retinol, which is then absorbed and stored in the liver in the ester form (especially retinyl palmitate; Fig. 126.2). Retinol and its esters represent the main dietary source, as well as transport and storage form, of vitamin A. Once inside the bloodstream, after release of the retinyl ester (storage form) from the liver, retinol is transported bound to a complex of retinol-binding protein (RBP) and transthyretin. The uptake of retinol from the RBP–transthyretin complex to target cells is mediated by STRA6 (stimulated by retinoic acid 6), a dimeric transmembrane protein that acts as an RBP–retinol receptor[4,5] (see Fig. 126.2).

The cellular pathway leading to conversion of retinol to a*t*-RA, the biologically active ligand that binds to nuclear RARs, consists of a two-step process. Retinol (vitamin A alcohol) is first reversibly oxidized to retinaldehyde (vitamin A aldehyde), which is then irreversibly converted to retinoic acid (vitamin A acid). Cellular RBP (CRBP-I) facilitates these enzymatic reactions by delivering retinol to appropriate enzymes (see Fig. 126.2).

Two additional intracellular carriers, cellular retinoic acid-binding proteins (CRABP-I and CRABP-II), are thought to transport retinoic acid to the nucleus and buffer the level of free a*t*-RA in the cell. The latter is accomplished by sequestering a*t*-RA and by promoting metabolism of a*t*-RA by cytochrome P450 enzymes belonging to the CYP26 family (e.g. CYP26A1). CRABP-I regulates the metabolic fate of its ligands by directly affecting the activities of RA-metabolizing cytochrome P450 enzymes, while CRABP-II markedly stimulates the RA-induced transcriptional activity of RAR through protein–protein interactions[6].

Intracellularly, retinoic acid is found in both all-*trans* and 9-*cis* configurations (see Fig. 126.2). Physiologically, a*t*-RA (tretinoin) is the predominant retinoic acid form, and only a small fraction is isomerized (enzymatically or non-enzymatically) into 13-*cis*-RA (isotretinoin) or converted to polar metabolites with unknown biologic activity. a*t*-RA is the primary active ligand that binds to the three known nuclear RARs,

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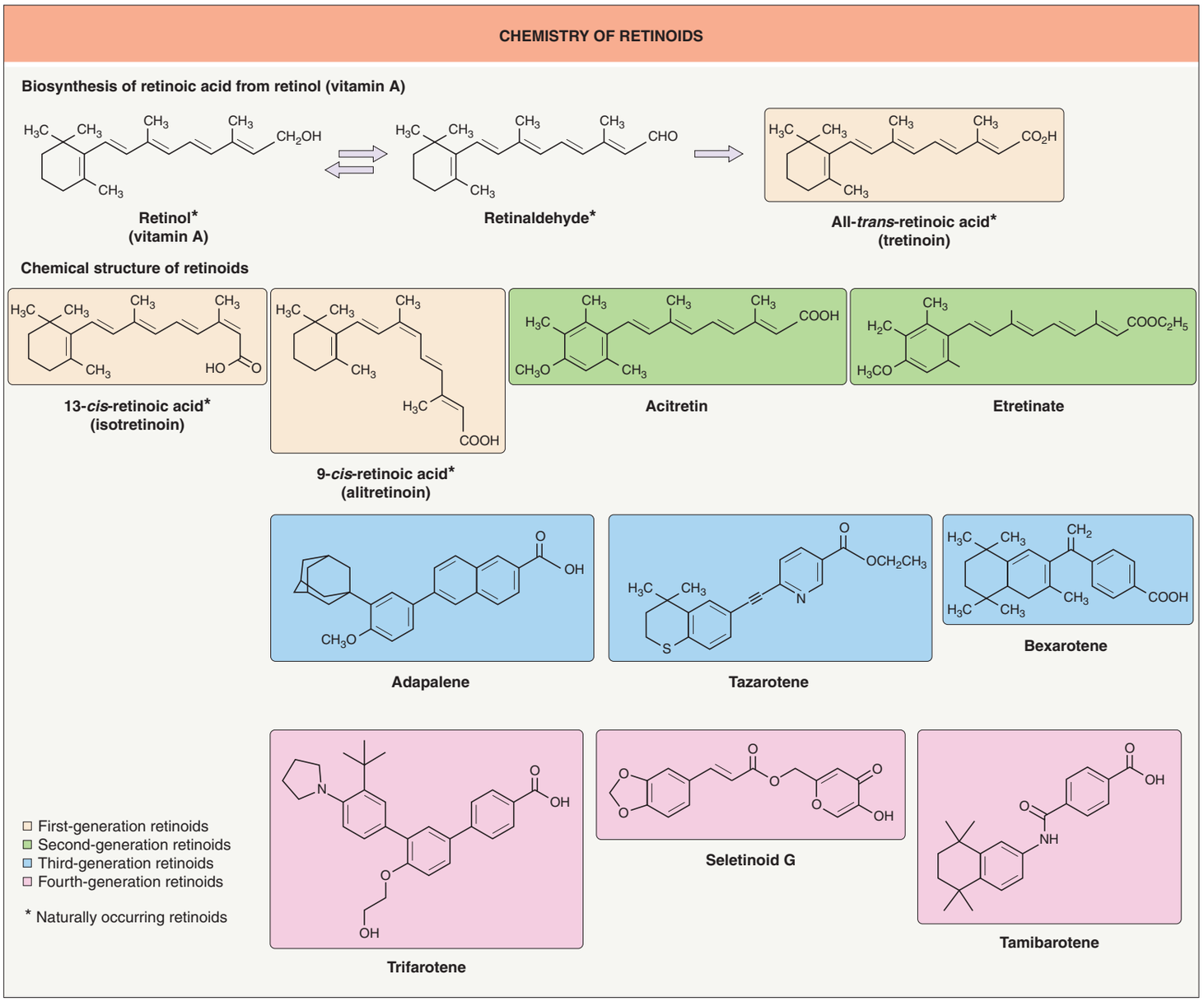

Fig. 126.1 Chemical structure of natural and synthetic retinoids. Motretinide is an additional second-generation retinoid.

BIOLOGIC FUNCTIONS OF RETINOIDS
Functions of vitamin A (retinol)*
• Embryonic growth • Morphogenesis • Differentiation and maintenance of epithelial tissues • Reproduction (retinol) • Visual function (retinaldehyde)
Biologic effects of retinoids
• Modulation of proliferation and differentiation • Anti-keratinization • Alteration of cellular cohesiveness • Anti-acne and anti-seborrheic effects • Immunologic and anti-inflammatory effects • Tumor prevention and therapy • Induction of apoptosis • Effects on extracellular matrix components
*Retinoic acid can substitute for retinol with regard to growth, morphogenesis, and epithelial differentiation; it cannot substitute completely for retinol in reproductive function, nor can it replace retinaldehyde in the visual system.

Table 126.1 Biologic functions of retinoids.

which mediate, at least in part, the molecular and cellular effects of retinoic acid. Although the biosynthesis of 9-*cis*-RA (alitretinoin) is not well-defined, it might be produced from all-*trans*-retinol via 9-*cis*-retinol and 9-*cis*-retinaldehyde.

Retinoid Receptors

Retinoids exert most of their physiologic effects on DNA transcription by binding to two distinct families of nuclear receptors, RARs and RXRs (see Fig. 126.2). These receptor families belong to a superfamily of nuclear receptors that act as ligand-activated transcription factors and include the steroid, vitamin D_3, and thyroid hormone receptors as well as peroxisome proliferator-activated receptors (PPARs). The RAR and RXR receptor families each contain three receptor isotypes (α, β and γ) encoded by different genes. a*t*-RA (tretinoin) binds only to RARs, whereas 9-*cis*-RA (alitretinoin) binds both to RARs and RXRs (Table 126.2).

RARs function as heterodimers with RXRs, whereas RXRs may also act as homodimers or participate in the formation of heterodimers with a variety of other nuclear receptors, including vitamin D_3, thyroid hormone, and PPARs. Such heterodimers provide a mechanism for cross-talk between nuclear hormone signaling pathways.

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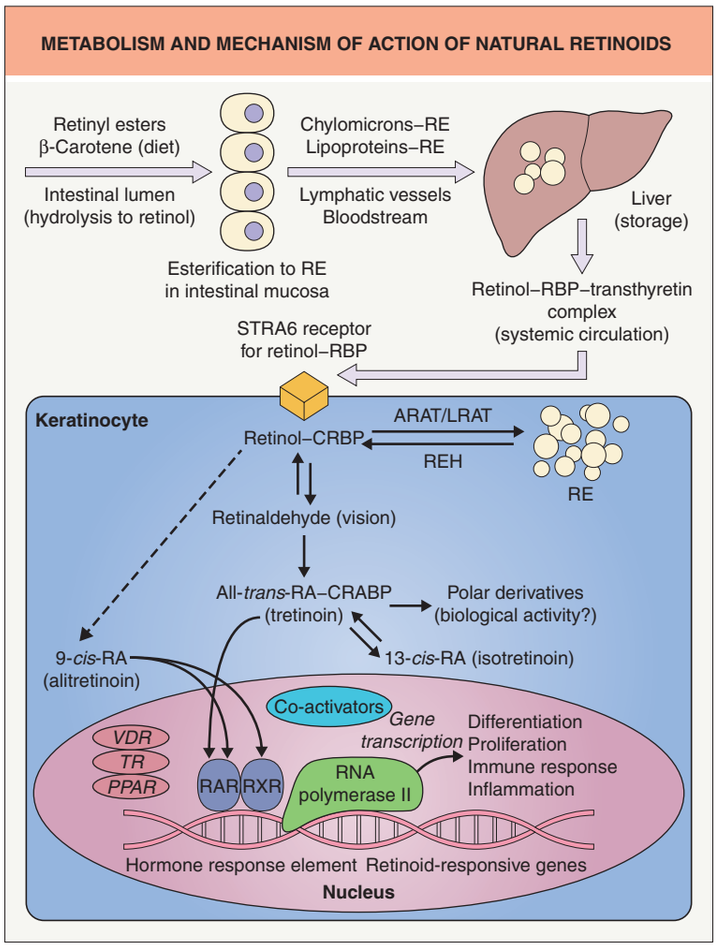

Fig. 126.2 Metabolism and mechanism of action of natural retinoids. ARAT, acyl-CoA:retinol acyl transferase; CRABP, cellular retinoic acid-binding protein; CRBP, cellular retinol-binding protein; LRAT, lecithin:retinol acyl transferase; PPAR, peroxisome proliferator-activated receptor; RA, retinoic acid; RAR, retinoic acid receptor; RBP, retinol-binding protein; RE, retinyl esters; REH, retinyl ester hydrolase; RXR, retinoid X receptor; STRA6, stimulated by retinoic acid 6; TR, thyroid receptor; VDR, vitamin D_3 receptor.

Dimers of retinoid receptors (RAR/RXR or RXR/RXR) are localized to the nucleus and bind, even in the unliganded state, to specific DNA regulatory sequences called retinoid hormone response elements (RAREs) in the promoter regions of retinoid-responsive genes (see Fig. 126.2). Unliganded receptors bind to co-repressor molecules and repress transcription. However, when the receptor binds its ligand, it undergoes a conformational change resulting in the release of co-repressors and recruitment of co-activators. These molecules include histone acetylases that change the conformation of chromatin and allow access to DNA by transcriptional machinery. The retinoid–receptor complex thereby modulates the transcription of specific sets of genes, and it is likely that many of the differentiation-inducing actions of retinoids are mediated by this mechanism.

In addition to *direct upregulation* of gene transcription via binding of retinoid–receptor complexes to RAREs, retinoids can have *indirect* effects on gene transcription that result from the *downregulation* of genes that do not contain RAREs in their promoter region. The retinoid–receptor complex is thought to antagonize transcription factors such as activator protein-1 (AP1) and nuclear factor-interleukin-6 (NF-IL6) by competing for the same set of required co-activator proteins, thereby leading to decreased expression of AP1 and NF-IL6-responsive genes; expression of the latter genes promotes cellular proliferation and inflammation. The antiproliferative and anti-inflammatory actions of retinoids are believed to be mediated by this type of indirect negative gene regulation. In particular, retinoic acid has important roles in inhibiting IL-6-driven induction of proinflammatory Th17 cells and in promoting differentiation of anti-inflammatory regulatory T cells (see Ch. 4). The anti-inflammatory effects of *at*-RA (tretinoin) have also been associated with its ability to regulate the expression and activation of Toll-like receptors (TLRs), especially TLR2. The biologic relevance of non-genomic effects of retinoids (e.g. cross-talk with extracellular protein/peptide signals acting via membrane receptors with downstream phosphorylation) remains under investigation.

The existence of several types and varied distributions of receptors, dimers, hormone response elements, and regulatory proteins shows that retinoid action is mediated via multiple pathways and results in a complex process of activation or inhibition of a large set of coordinately regulated genes. However, the exact mechanism of retinoid action in many dermatologic conditions is still unknown, and, so far, the nuclear receptor concept does not fully explain the biologic diversity of retinoid effects.

Retinoids in the Skin

Vitamin A and its bioactive metabolites play a major role in promoting the keratinocyte differentiation that occurs from the epidermal basal layer to the stratum corneum[7]. Retinyl esters and retinol are the major retinoids found in the skin, with the former reaching a concentration of >1000 pmol/g in the epidermis. Epidermal retinoids can absorb UVB radiation and have ~2%–3% of the UVB-filtering capacity of the melanin in phototype II/III skin[7,8]. The UVB absorption process results in dramatic depletion of epidermal retinoids[9], which can be prevented by application of topical retinoids[10]. Nutritional deficiency, aging, and oxidative stress can also lead to lowered retinoid levels in the skin[7]. There has been ongoing debate regarding possible generation of reactive oxygen species upon the interaction of UVA radiation with retinyl esters (e.g. retinyl palmitate) present in cosmetics and sunscreens, but the theoretical potential for photocarcinogenicity has not been shown to be clinically relevant.

Synthetic Retinoids

There are now four generations of retinoids (see Fig. 126.1). The *first-generation* non-aromatic retinoids (tretinoin, isotretinoin, alitretinoin) are synthesized by chemically modifying the polar end group and the polyene side chain of vitamin A. *Second-generation* mono-aromatic retinoids (etretinate, acitretin, motretinide) are formed by replacing the cyclic moiety of vitamin A with different substituted and unsubstituted ring systems. *Third-generation* polyaromatic retinoids (adapalene, bexarotene, tazarotene), called arotinoids, are produced by cyclization of the polyene side chain. *Fourth-generation* polyaromatic retinoids (trifarotene, seletinoid G, tamibarotene) were designed to have increased selectivity of binding.

Commercially available retinoids differ not only in their spectrum of clinical efficacy but also in their observed toxicity and pharmacokinetics. Each retinoid should be investigated as a unique drug, and the lack of disease response to one retinoid does not necessarily signify unresponsiveness to other retinoids.

The oral bioavailability of retinoids is considerably enhanced when administrated with food, especially with a fatty meal, due to their lipophilic properties. More recently, a lidose–isotretinoin formulation with enhanced absorption when taken without food was developed. Retinoid metabolism is predominantly hepatic; it involves oxidation and chain shortening to biologically inactive and polar metabolites, facilitating biliary and/or renal elimination. The oxidative metabolism is induced by the retinoids themselves and possibly also by other agents known to stimulate hepatic cytochrome P450 isoforms. Table 126.2 summarizes the key pharmacologic features and nuclear binding profiles of retinoids used in dermatology.

First-generation retinoids

Tretinoin (all-*trans*-retinoic acid)

Tretinoin, the predominant physiologic form of retinoic acid, was the first topical retinoid to be utilized for the treatment of acne and photoaging. It binds to cellular retinoic acid-binding proteins (CRABPs) and non-selectively to nuclear RARs (see above), resulting in modulation of keratinocyte proliferation and differentiation, inflammation, and extracellular matrix production. Because the standard tretinoin formulation is photolabile, night-time application is recommended to prevent degradation.

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KEY PHARMACOLOGIC FEATURES AND NUCLEAR BINDING PROFILES OF RETINOIDS												
Retinoid	Type	Systemic absorption or bioavailability* (% dose)	Elimination half-life	Metabolism	Excretion	Nuclear receptor binding affinity						
						RAR			RXR			
						α	β	γ	α	β	γ	
Topical retinoids*												
Tretinoin† (all-*trans*-retinoic acid)	Natural (1st generation)	<2% in normal skin	Normally present in the skin	Isomerization to 13-*cis*-RA (epidermis)	Desquamation, biliary	+	+	+	–	–	–	
Alitretinoin (9-*cis*-retinoic acid)	Natural (1st generation)	Minimal	Normally present in the skin	Main metabolite: 4-oxo-9-*cis*-RA	Desquamation, biliary	+	+	+	+	+	+	
Adapalene	3rd generation	Minimal	Unknown	Minimal biotransformation due to chemical rigidity	Desquamation, biliary	±	+	+	–	–	–	
Tazarotene	3rd generation	<1%–6% in normal skin	16 hours (tazarotenic acid)	Rapid (within 20 minutes) ester hydrolysis within the skin to form its active metabolite, tazarotenic acid; excreted metabolites: sulfoxides, sulfones	Desquamation, renal (within 3 days), biliary (within 7 days)	±	+	+	–	–	–	
Trifarotene	4th generation	≤10 pmol/L	2–9 hours	Hepatic	Biliary	–	–	+	–	–	–	
Systemic retinoids												
Tretinoin†	Natural (1st generation)	50%	1 hour	Hepatic; main metabolites: *cis*- and *trans*-4-oxo derivatives	Biliary, renal	+	+	+	–	–	–	
Isotretinoin (13-*cis*-RA)	Natural (1st generation)	25%	20 hours	Hepatic, endogenous concentration reached within 2 weeks; main metabolite: 4-oxo-isotretinoin	Biliary, renal	–	–	–	–	–	–	
Alitretinoin	Natural (1st generation)	Unknown	2–10 hours	Hepatic; main metabolite: 4-oxo-alitretinoin	Renal, biliary	+	+	+	+	+	+	
Acitretin (and etretinate)	2nd generation	60%	2 days	Hepatic, re-esterification to etretinate indirectly increased by alcohol consumption; main metabolite: *cis*-acitretin	Biliary, renal	+‡	+‡	+‡	–	–	–	
Bexarotene¶	3rd generation	Unknown	7–9 hours	Hepatic	Biliary	–	–	–	+	+	+	
Tamibarotene	4th generation	NA	2–3 hours	Hepatic	NA	+	+	–	±	±	±	

*Topical formulations have limited propensity for distribution to systemic tissues; oral bioavailability is highly variable and increases when administered with food.
†Also binds the cellular retinoic acid-binding protein (CRABP).
‡Activate RARs despite binding poorly to them.
¶Also available in a topical form.

Table 126.2 Key pharmacologic features and nuclear binding profile of retinoids. NA, not available; RA, retinoic acid; RAR, retinoic acid receptor; RXR, retinoid X receptor.

Isotretinoin (13-*cis*-retinoic acid)

Isotretinoin is a naturally occurring retinoid resulting from the metabolism of vitamin A (see Fig. 126.2)[11]. 13-*cis*-RA and a*t*-RA are two physiologically interconvertible isomers that differ in their elimination half-lives: approximately 20 hours and 1 hour, respectively. Isotretinoin undergoes first-pass metabolism in the liver and subsequent enterohepatic recycling. In plasma, more than 99% of isotretinoin is protein-bound, mainly to albumin. There is neither liver nor adipose tissue storage, in contrast to vitamin A. The major metabolite of isotretinoin, 4-oxo-isotretinoin, is produced by oxidation. Isotretinoin and its metabolites are excreted in the urine and feces. After discontinuation of isotretinoin, *physiologic* concentrations of the drug and its major metabolites are reached within 2 weeks, ranging from 2 days for a*t*-RA to 10 days for 4-oxo-isotretinoin. Therefore, 1 month of post-therapy contraception provides an adequate safety margin to avoid teratogenicity.

Among natural and synthetic retinoids, only oral isotretinoin and its 4-oxo-isotretinoin metabolite significantly suppress sebum production, which explains their dramatic effects on acne. Isotretinoin-induced seboatrophy is unlikely to be RAR/RXR-mediated since isotretinoin has no clear affinity for any known retinoid receptor[12], and potent RAR/RXR ligands such as tretinoin and alitretinoin do not lead to seboatrophy. The mechanism of isotretinoin-induced seboatrophy is thought to involve sebocyte apoptosis[13], but inhibition of either sebocyte progenitor differentiation or the genes encoding sebogenic enzymes represent two other possible pathomechanisms.

Alitretinoin (9-*cis*-retinoic acid)

Alitretinoin is a naturally occurring pan-agonist retinoid that binds to both RARs and RXRs[14], representing the endogenous ligand for the latter. Topical alitretinoin 0.1% gel is FDA-approved for the treatment of cutaneous Kaposi sarcoma, and oral alitretinoin therapy is currently utilized outside the US for refractory chronic hand eczema. The absorption of oral alitretinoin from the gastrointestinal tract is variable and dose-proportional over the therapeutic range of 10–30 mg daily; the absolute bioavailability of alitretinoin has not been determined. Alitretinoin is metabolized via oxidation by CYP3A4 enzymes of the liver into 4-oxo-alitretinoin, and both the parent drug and its metabolite also undergo isomerization into a*t*-RA and 4-oxo-a*t*-RA, respectively. Elimination is primarily renal, with the half-life of unchanged alitretinoin ranging between 2 and 10 hours. Alitretinoin concentrations return to normal within 1–3 days after treatment cessation.

The mechanism of action of alitretinoin in chronic hand eczema is unknown, although downregulated expression of RARs and RXRs in lesional skin was recently described[15]. Alitretinoin has immunomodulatory and anti-inflammatory effects that include suppressing the expansion of cytokine-activated leukocyte subsets and antigen-presenting cells.

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Second-generation retinoids

Acitretin

Acitretin is the major metabolite and the pharmacologically active form of etretinate[16]. Although acitretin and etretinate are equally effective, acitretin has much more rapid elimination. Etretinate is approximately 50 times more lipophilic than acitretin and binds strongly to plasma proteins, particularly lipoproteins and albumin. Etretinate is stored in adipose tissue (including subcutaneous fat), from which it is released slowly, with a long terminal half-life of up to 120 days. Under identical conditions, acitretin carries a negatively charged group that makes it much less lipophilic than etretinate. As a result, acitretin does not accumulate in adipose tissue and is eliminated from the body more rapidly, with a half-life of 2 days.

The major serious adverse effect of oral retinoids is teratogenicity, and, therefore, the length of time that these drugs are present in the body is of great importance. Re-esterification of acitretin to etretinate may occur when acitretin is taken simultaneously with alcohol[17,18]. This finding prompted the manufacturer to extend the time of contraception for female patients with childbearing potential taking acitretin to 2 years after its discontinuation, as it is for etretinate. The FDA advises a contraceptive period of at least 3 years for acitretin, based on the pharmacokinetics of acitretin and etretinate observed in clinical trials and on previous safety experience with etretinate. However, acitretin retains its shorter half-life in women who strictly avoid alcohol during treatment and for 2 months thereafter.[16] Because of its improved risk–benefit ratio, acitretin was approved by the FDA in 1996 as a substitute for etretinate, which is no longer commercially available[16].

Acitretin's mechanism of action is still not clearly understood. Paradoxically, it activates all three RAR subtypes despite binding poorly to them. Evidence supports a normalization of differentiation and proliferation as well as a modification of inflammatory responses and neutrophil function.

Third-generation retinoids

Adapalene

Adapalene is a photostable, rigid, and highly lipophilic synthetic retinoid with higher affinity for RAR-β/γ than for RAR-α. Since RAR-β is not expressed in keratinocytes, RAR-γ is the primary target receptor for adapalene in the epidermis. Adapalene does not bind to cellular retinoic acid-binding proteins (CRABPs) but does induce expression of CRABP-II. Its lipophilic properties may contribute to better pilosebaceous uptake and anti-acne activity. Given its negligible transdermal absorption, the teratogenic risks of topical adapalene appear to be minimal. Adapalene affects the cellular differentiation, keratinization, and inflammatory processes that are abnormal in acne. Although adapalene has not been shown to be sebostatic *in vivo* in humans, it was found to inhibit triacylglycerol biosynthesis and lipid droplet accumulation *in vitro* in hamster sebocytes[19].

Tazarotene

Tazarotene is a prodrug that is rapidly converted by cutaneous esterases to its free carboxylic acid (tazarotenic acid), which is the active metabolite. It has a higher affinity for RAR-β/γ than RAR-α and no affinity for RXR[20]. Because of its rapid metabolism, systemic exposure is low. Tazarotene appears to modulate the pathogenesis of psoriasis by regulating expression of retinoid responsive genes, including those involved in cell proliferation, cell differentiation, and inflammation[21].

Bexarotene

Bexarotene, an RXR-selective retinoid (rexinoid), is FDA-approved in oral (1999) and topical (2000) forms for treatment of CTCL. This compound is about 100-fold more potent at binding to RXRs than to RARs. In plasma, bexarotene is highly bound (>99%) to various proteins that have yet to be characterized. Bexarotene has a clearance profile similar to that of isotretinoin, with a terminal half-life of between 7 and 9 hours[22]. Bexarotene is metabolized by CYP3A4 and generates its own oxidative metabolites via hepatic CYP3A4 induction. Neither bexarotene nor its metabolites are excreted in the urine; their elimination is thought to occur primarily via the hepatobiliary system[22]. The exact mechanism of action of bexarotene in CTCL is still unknown, but it probably acts through regulation of cellular differentiation and proliferation, as well as via induction of apoptosis[23].

Fourth-generation retinoids

Trifarotene

Trifarotene, a fourth-generation retinoid with 20-fold higher affinity for RAR-γ than RAR-α/β, was FDA-approved in 2019 as a topical acne treatment for individuals ≥9 years of age[24]. This agent was designed to be stable in keratinocytes but rapidly metabolized in hepatic microsomes, thereby minimizing systemic drug levels. In addition to effects on known retinoid-mediated pathways, gene expression profiling has shown that trifarotene modulates proteolysis, reduces intercellular adhesion, and activates transporters involved in skin hydration[25].

Retinoic acid metabolism blocking agents

*R*etinoic *a*cid *m*etabolism *b*locking *a*gents (RAMBAs) inhibit catabolism of a*t*-RA by P450 enzymes in the CYP26 family, thereby raising the intracellular levels of endogenous a*t*-RA in the skin and other targeted tissues[26]. This limits systemic exposure and the potential for toxicity, and a*t*-RA levels return to the physiologic range within 24 hours of discontinuing these medications. Liarozole, an imidazole derivative without antifungal properties, is a RAMBA with low isoenzyme specificity that also inhibits P450-mediated pathways of steroid biosynthesis whereas talarozole, a triazole derivative, is a more selective and highly active CYP26 enzyme inhibitor. DX314, a CYP26B1-selective RAMBA that may protect epidermal barrier integrity, is also under investigation[27]. Liarozole has been granted orphan drug status for congenital ichthyosis by the European Commission and the FDA. In a randomized, placebo-controlled phase II/III trial in 64 patients with moderate-to-severe lamellar ichthyosis, a 41%–50% response rate was observed in patients receiving liarozole, compared to 11% with placebo, with borderline statistical significance ($p = 0.056$)[26].

INDICATIONS (TABLE 126.3)

Topical Retinoids

The most important element in topical retinoid therapy is patient education. Local skin irritation can be expected, and noticeable beneficial effects often take weeks or months to appear. Administration of topical retinoids should be titrated depending on cutaneous irritation, which may mean decreasing the concentration or the frequency of application. It is generally wise to begin with a low-strength formulation and then increase the concentration as tolerance builds. Another tactic is to start by applying a given concentration every other day. Daytime moisturizers with sunscreen are important components of any topical retinoid regimen.

Acne

Topical retinoids, of which tretinoin is the prototype, are mainstays in the treatment of comedonal and inflammatory acne vulgaris[28]. The primary mode of action in acne is thought to be the normalization of differentiation and proliferation of the follicular epithelium (see Ch. 36), which leads to the loosening and unseating of microcomedones, thereby preventing obstruction of the pilosebaceous unit. In addition, topical retinoids may have anti-inflammatory activity. The mechanism of action of topical isotretinoin (not available in the US) is similar to that of topical tretinoin because of intraepithelial isomerization of isotretinoin to tretinoin (see Fig. 126.2). In contrast to oral isotretinoin, topical isotretinoin fails to suppress sebum production; it is less irritating and probably also somewhat less effective than tretinoin. The topical synthetic retinoids adapalene, tazarotene, and trifarotene have comparable or (for adapalene) improved tolerability while maintaining efficacy similar to that of topical tretinoin[29,30].

Topical retinoids should be applied to the entire face or all acne-prone areas once a day as tolerated, with tretinoin applied in the evening to prevent inactivation by UV light. Application of these medications to dry skin may reduce skin irritation by minimizing dermal absorption. Patients should be advised that beneficial effects do not typically become evident for 6–8 weeks or longer. An apparent exacerbation may occur during the first month of therapy, representing the externalization of deeper-seated acne lesions as the follicular epithelium is loosening. With the exception of mild, primarily comedonal acne, topical retinoids are typically used concomitantly with benzoyl peroxide products and antibiotics (oral or topical) that target *Cutibacterium acnes* and inflammatory processes that contribute to acne (see Ch. 36).

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CLINICAL INDICATIONS FOR RETINOIDS IN DERMATOLOGY		
Topical retinoids		
FDA-approved indications	Acne vulgaris (tretinoin, adapalene, tazarotene, trifarotene) Photoaging, e.g. fine facial wrinkling, mottled pigmentation, facial roughness (tretinoin, tazarotene) Psoriasis, <20% of body surface area (tazarotene*) Cutaneous T cell lymphoma (bexarotene) Kaposi sarcoma (alitretinoin)	
Selected non-FDA-approved indications	Some disorders of cornification (ichthyoses) Darier disease Pityriasis rubra pilaris Rosacea Pigmentary disorders, e.g. melasma, postinflammatory hyperpigmentation Lentigines Actinic keratoses Striae Wound healing Lichen planus (oral and cutaneous) Verrucae planae (flat warts) Corticosteroid-induced atrophy Therapy and prevention of skin cancer (basal cell carcinoma; xeroderma pigmentosum)	
Systemic retinoids		
FDA-approved indications	Psoriasis (acitretin)	Pustular psoriasis (localized and von Zumbusch) Erythrodermic psoriasis Severe and recalcitrant plaque-type psoriasis*
	Acne (isotretinoin)	Nodulocystic acne Recalcitrant acne with tendency for scarring
	Cutaneous T cell lymphoma (bexarotene)	Resistant to at least one systemic therapy
Selected non-FDA-approved indications	Rosacea and acne-related disorders	Hidradenitis suppurativa Eosinophilic pustular folliculitis (Ofuji disease) AIDS-associated eosinophilic folliculitis Pyoderma faciale (rosacea fulminans) Acne with solid facial edema Dissecting cellulitis of the scalp
	Disorders of cornification and related conditions	Ichthyoses, various forms (see Ch. 57) Darier disease Keratoderma Papillon–Lefèvre syndrome
	Inflammatory disorders	Chronic hand eczema (alitretinoin, acitretin) Pityriasis rubra pilaris Lupus erythematosus Lichen planus
	Chemoprophylaxis of neoplastic processes	Xeroderma pigmentosum Basal cell nevus syndrome Skin cancer in solid organ transplant patients
	Treatment of neoplastic processes	Epithelial precancerous conditions Basal cell carcinoma Advanced squamous cell carcinoma Keratoacanthoma (including those induced by BRAF inhibitors and immune checkpoint inhibitors) Kaposi sarcoma Sebaceous hyperplasia Muir–Torre syndrome Leukoplakia Langerhans cell histiocytosis
	Miscellaneous conditions	Atrophoderma vermiculatum Ulerythema ophryogenes Confluent and reticulated papillomatosis Bazex syndrome (paraneoplastic acrokeratosis) Follicular mucinosis Sarcoidosis Granuloma annulare Lichen sclerosus (genital or extragenital) Subcorneal pustular dermatosis Graft-versus-host disease Verruca vulgaris, condyloma
*Combination therapy, e.g. with a topical corticosteroid or photo(chemo)therapy, is often required.		

Table 126.3 Clinical indications for retinoids in dermatology. Of note, the oral form of tretinoin (all-*trans*-retinoic acid; *at*-RA) is FDA-approved for the treatment of acute promyelocytic leukemia (APL). Tamibarotene, a fourth-generation retinoid, is currently approved in Japan for the treatment of APL.

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Psoriasis

Topical application of tretinoin or isotretinoin has limited efficacy in psoriasis. Although topical tazarotene was shown to be effective in treating mild to moderate plaque-type psoriasis affecting ≤20% of the body surface area, irritation limits its use as monotherapy. A combination of tazarotene and a mid- to high-potency topical corticosteroid can improve efficacy and reduce the likelihood of both irritation and corticosteroid-induced atrophy.

Photoaging

Several controlled studies have demonstrated that topical retinoids, including tretinoin and tazarotene creams, can improve fine wrinkling and lighten uneven pigmentation[31]. It generally takes 3–6 months of daily application to see significant clinical improvement. Cutaneous irritation is usually the limiting factor. Retinaldehyde was found to be as effective as tretinoin in treating photodamage in less extensive trials[32], and it had a better tolerance profile that may reflect more controlled delivery of retinoic acid to target cells. Seletinoid G, a fourth-generation polyaromatic retinoid with RAR-γ selectivity designed to produce less skin irritation, is under investigation as a treatment for photoaging.[33]

Photoaging is the consequence of UVR-induced damage to the skin, which is characterized by decreased vitamin A content[7] as well as lower expression of RXR-α and RAR-γ (the major retinoid receptors in keratinocytes)[34] in the acute setting and then later by upregulation of AP1-driven matrix metalloproteinases (MMPs). Topical retinoids promote cellular differentiation and extracellular matrix synthesis, including an increase in hyaluronic acid via a CD44-mediated mechanism in addition to decreased MMP production[35]. Histologic findings after repeated topical application of tretinoin include compaction of the stratum corneum, epidermal hyperplasia (acanthosis), correction of atypia (e.g. actinic keratoses), dispersion of melanin granules, increased dermal collagen synthesis, and angiogenesis. These findings explain reported observations of smoother skin, a rosy glow, decreased blotchy pigmentation, and diminished fine lines and wrinkles following topical retinoid treatment[36].

Other indications

Other FDA-approved indications for topical retinoids include alitretinoin 0.1% gel for cutaneous Kaposi sarcoma and bexarotene 1% gel for CTCL[37]. There are also numerous "off-label" uses for topical retinoids (see Table 126.3).

Systemic Retinoids

Psoriasis

Several randomized, controlled, multicenter trials have been performed to ascertain the efficacy of acitretin in plaque-type and pustular psoriasis, with similar results to prior studies of etretinate[38]. Guidelines for use of acitretin in psoriasis are summarized in Tables 8.13 and 8.14. The best results have been obtained in acral or generalized (von Zumbusch) pustular psoriasis, in which acitretin is considered to be a first-line therapy. The various forms of pustular psoriasis, as well as erythrodermic psoriasis, tend to clear relatively rapidly with acitretin monotherapy. Rebound does not usually occur after stopping treatment, and reintroduction produces a beneficial response.

Plaque-type psoriasis responds variably to acitretin. Although complete clearance of plaques is achieved in only about 30% of treated patients, substantial improvement is obtained in an additional 50%. The mean decrease in the Psoriasis Area and Severity Index (PASI) score is ~60%–70%, depending on the dosage and duration of treatment. Many of the plaques may remain, but they are thinner with less scale and erythema.

An initial worsening with increased erythema and/or extent of involvement of plaque-type psoriasis may occur within a few days of starting acitretin therapy at a dosage of 0.5–1 mg/kg per day (~25–70 mg/day in adults). Utilization of a lower dose (10 mg/day) initially, followed by a progressive increase, helps to avoid this complication.

Total clearance of psoriatic plaques usually requires that oral retinoids be combined with other treatments such as topical corticosteroids, topical vitamin D derivatives, anthralin (dithranol), or photo(chemo)therapy (narrowband UVB or psoralen plus UVA [PUVA])[39,40]. The "Re-PUVA" combination, in which retinoids are given for 14 days before starting PUVA, can produce an accelerated response rate and resolution of lesions that would not clear with a retinoid or PUVA alone (Fig. 126.3). The cumulative UVA exposure required to produce a remission is also significantly lower with Re-PUVA, reducing the risk of UVA-induced carcinogenesis. Oral retinoids can also be combined with bath PUVA. Lastly, the combination of acitretin with UVB phototherapy has been shown to be more effective than retinoid or UVB phototherapy alone in patients with psoriasis[41,42].

Isotretinoin is less effective than acitretin for psoriasis, although some efficacy has been shown in combination with PUVA[43]. Isotretinoin is still occasionally used in women of childbearing age with psoriasis who need a systemic retinoid to avoid the long post-acitretin contraception period.

Acne

The introduction of oral isotretinoin was a key advance in acne therapy and it still remains the most effective medication. With the possible exception of hormonal acne, isotretinoin has been shown to induce long-term remissions and even "cure" acne. It is the only medication that affects, albeit not to the same degree or permanently, all the major etiologic factors implicated in acne: sebum production, comedogenesis, and colonization with *C. acnes* (Fig. 126.4; see Ch. 36). Among natural and synthetic retinoids studied in humans, only isotretinoin has been found to suppress sebum production (see above).

In the early 1980s, oral isotretinoin use was restricted to patients suffering from severe nodulocystic acne. With increasing experience, however, its use has been extended to patients with less severe disease who have responded unsatisfactorily to conventional therapies such as topical retinoids plus oral antibiotics. Patients with moderate acne that shows signs of scarring also represent candidates for oral isotretinoin therapy.

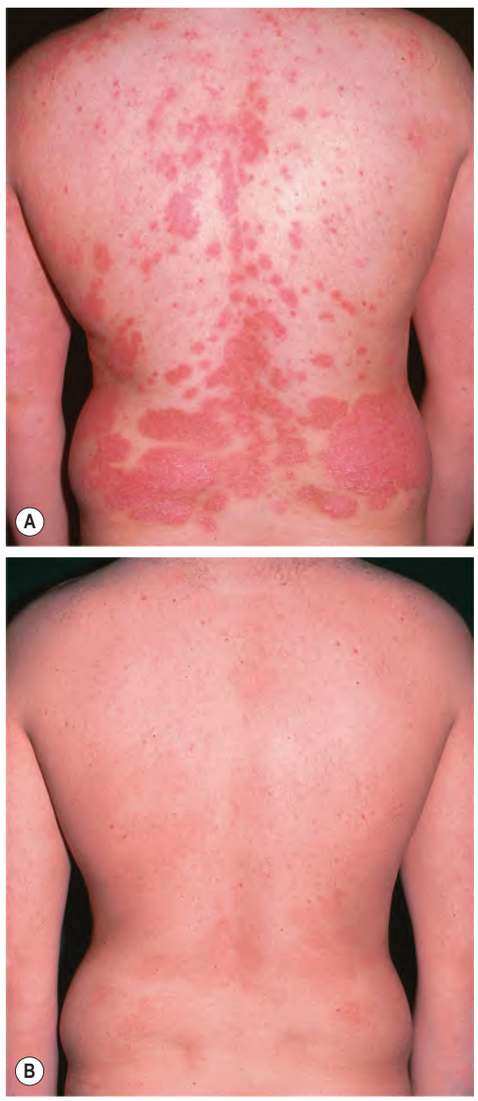

Fig. 126.3 Chronic plaque-type psoriasis. Before (**A**) and after (**B**) treatment with a combination of acitretin and PUVA.

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Fig. 126.4 Severe acne vulgaris. Before (**A**) and after (**B**) treatment with oral isotretinoin. *Courtesy Ángela Hernández-Martín, MD.*

Acne morbidity extends beyond the aesthetics of physical appearance. Psychosocial consequences of acne range from depression and anxiety to interpersonal and work-related difficulties. Quality-of-life studies have shown that isotretinoin treatment significantly improves sociability and self-esteem.

It was initially considered that optimal benefit would be achieved with an isotretinoin dose of 1 mg/kg per day. However, this can induce undesirable dose-dependent effects, and similar short-term therapeutic results have been obtained with doses below 0.5 mg/kg per day. In order to avoid the higher incidence of relapses associated with lower-dose isotretinoin, treatment can be continued for a longer period of time in order to reach a critical cumulative dose threshold. The cumulative dose was first recognized to be important in isotretinoin therapy for acne by Harms et al. in 1989[44]; it refers to the total amount of oral isotretinoin taken by the patient over the entire course, divided by the body weight (mg/kg body weight). For example, a patient weighing 50 kg and receiving 25 mg/day of isotretinoin for 100 days would have received a cumulative dose of 50 mg/kg (25 mg × 100 = 2500 mg, divided by 50 kg = 50 mg/kg). Data from several studies indicated that post-therapy relapse is minimized by a treatment course amounting to a total of at least 120 mg/kg, with minimal further therapeutic gain beyond about 150 mg/kg[45]. However, a 6-month course of low-dose isotretinoin (20 mg/day; mean cumulative dose = 70 mg/kg) was also found to be effective in the treatment of moderate acne, with a low incidence of severe side effects and a reduced cost compared with higher doses[46].

A lag period of 1–3 months may occur before the onset of the therapeutic effect. Approximately one-third of patients with acne require a second course of isotretinoin therapy, either for persistent disease or for relapse. A major predictive factor for resistance to isotretinoin treatment is closed comedonal and microcystic acne. Hyperandrogenemia due to ovarian or adrenal dysfunction should be excluded in women with a history of acne that is either refractory to isotretinoin or has relapsed following appropriate courses of isotretinoin.

A flare of disease during the first few weeks of treatment, sometimes with an acne fulminans-like presentation and subsequent evolution of acne cysts into lesions resembling pyogenic granulomas, is occasionally observed with isotretinoin treatment (see Fig. 36.17). The incidence of this side effect may be reduced by using lower doses of isotretinoin during the first 3–4 weeks of therapy[47]. Because oral isotretinoin is a potent teratogen, prevention of pregnancy in female patients with childbearing potential is critical, with a need for continuous effective contraception from 1 month prior to initiating therapy until 1 month after the last dose (see below)[47].

Isotretinoin has a more limited effect on hidradenitis suppurativa and dissecting cellulitis of the scalp, which can occur together with acne conglobata as components of the follicular occlusion tetrad[48]. Some clinicians recommend oral isotretinoin during the weeks or months preceding surgery for hidradenitis suppurativa, and sometimes also during the postoperative period, while others avoid isotretinoin perioperatively because of reports of enhanced scarring (see below).

Cutaneous T cell lymphoma

Isotretinoin and acitretin are somewhat effective and are considered to be of equal potency in the treatment of mycosis fungoides (MF). Retinoids can be used in combination with PUVA, interferon, or systemic chemotherapy (see Ch. 120). Despite initial improvement of the various stages of MF, it is often impossible to maintain remissions with these retinoids as monotherapy.

Both oral and topical bexarotene can produce clinical responses in CTCL. Two multicenter clinical trials of oral bexarotene were conducted in patients with stage I–IIA (early) disease (n = 58) or stage IIB–IVB (advanced) disease (n = 94)[23]. Patients in the early-stage study randomly assigned to a low-dose arm of 6.5 mg/m^2 per day had a 20% response rate, in contrast to 54% and 67% for patients receiving starting doses of 300 mg/m^2 per day and 650 mg/m^2 per day (then decreased to 500 mg/m^2 per day), respectively[41]. However, dose-dependent toxicity was a significant problem. Response rates were ~50% for the patients with advanced CTCL, who all received doses of 300–650 mg/m^2 per day[23,37].

Bexarotene is FDA-approved as oral therapy for CTCL that has been refractory to at least one systemic agent, and as a topical gel formulation for refractory early-stage CTCL. Given the high incidence of side effects, in particular hyperlipidemia (see below), a commonly recommended starting dose for oral bexarotene is 150 mg/m^2 per day (Table 126.4). Combination therapies utilizing bexarotene together with PUVA, narrowband UVB, methotrexate, interferon-α, or denileukin diftitox have also been evaluated in clinical trials and case series[49].

Chronic hand eczema

In a randomized, double-blind, placebo-controlled, multicenter study, 1032 adults with severe chronic hand eczema were treated with 30 mg or 10 mg of oral alitretinoin versus placebo once daily for 12–24 weeks. Responses, defined as clear or almost clear hands, were achieved in 48%, 28%, and 17% of the patients in the three groups, respectively. Treatment was well tolerated, with dose-dependent adverse events including headache, mucocutaneous side effects, hyperlipidemia, and hypothyroidism. Improvement typically began to develop after ~4 weeks of treatment, and the median time to relapse, defined as recurrence of 75% of initial signs and symptoms, was 6 months in the absence of any further therapy[50]. Oral alitretinoin (currently not available in the US) therefore represents a therapeutic option for adults with severe, recalcitrant chronic hand eczema, with the possibility of using intermittent courses to induce intermediate-term remissions[51].

Other "off-label" clinical uses

The multiple effects of retinoids explain their role in treating a broad spectrum of cutaneous diseases. Although many skin disorders respond to retinoids (see Table 126.3), benefit has been established in controlled studies for relatively few conditions.

Ichthyosis

Acitretin and isotretinoin can be of benefit in patients with non-syndromic autosomal recessive congenital ichthyoses (Fig. 126.5; see Ch. 57)[52]. Good results have also been observed in patients with X-linked recessive ichthyosis and ichthyosis vulgaris; however, these diseases tend to be less severe and do not usually require systemic retinoid therapy. Given the lifelong duration of heritable ichthyoses, intermittent retinoid courses are sometimes prescribed.

Darier disease

Severe forms of Darier disease are often treated with systemic retinoids. Therapy should be started with a low dose (e.g. acitretin 10 mg/day,

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APPROACH TO ORAL BEXAROTENE THERAPY FOR CUTANEOUS T-CELL LYMPHOMA			
	Bexarotene administration	**Monitoring**	**Management of side effects**
Prior to treatment		Baseline CBC, basic metabolic panel, hepatic panel, TSH, free T4, fasting lipid panel, CK Determine if patient takes medications that inhibit or induce CYP3A4 (see Ch. 131)	Dietary and lifestyle modifications to minimize risk of hyperlipidemia If pre-existing laboratory abnormalities, administer: • Levothyroxine to achieve normal free T4 levels • A fibrate (e.g. fenofibrate)* to achieve triglyceride level <200 mg/dl and HDL level >50 mg/dl • A statin† to achieve LDL level <160 mg/dl
Day –7			Start fibrate (e.g. fenofibrate 145–200 mg daily)*
Day 0	Start bexarotene 150 mg/m^2 daily		Start levothyroxine 0.05 mg daily
Days 7 to 28		Free T4‡ & fasting lipid panel every 1–2 weeks	Adjust levothyroxine and fibrate doses and/or add statin† as required to keep laboratory values in target ranges above
Day 28¶	Increase to bexarotene 300 mg/m^2 daily		
Day 29 onward	Adjust bexarotene dose as needed depending on disease response	Free T4‡ & fasting lipid panel • Every 1–2 weeks until in target range and after subsequent dose changes • Otherwise every 1–2 months CBC every 1–2 months	As above; a triglyceride level <400 mg/dl may be a more realistic target for some patients

*Reduce dose for patients with decreased creatinine clearance; because gemfibrozil raises bexarotene levels, these medications should ***not*** be administered concomitantly.
†Niacin represents another option for patients with combined hyperlipidemia; patients receiving a fibrate and statin concomitantly may be at increased risk of rhabdomyolysis.
‡Monitor free T4 as indicator of hypothyroidism; bexarotene lowers TSH levels (central hypothyroidism).
¶May increase dose as early as day 14 or, if more time is required to normalize lipid and free T4 levels, after day 28.

Table 126.4 Approach to initiating oral bexarotene therapy for cutaneous T cell lymphoma. CBC, complete blood count; CK, creatine kinase; HDL, high-density lipoprotein; LDL, low-density lipoprotein; T4, thyroxine; TSH, thyroid-stimulating hormone.

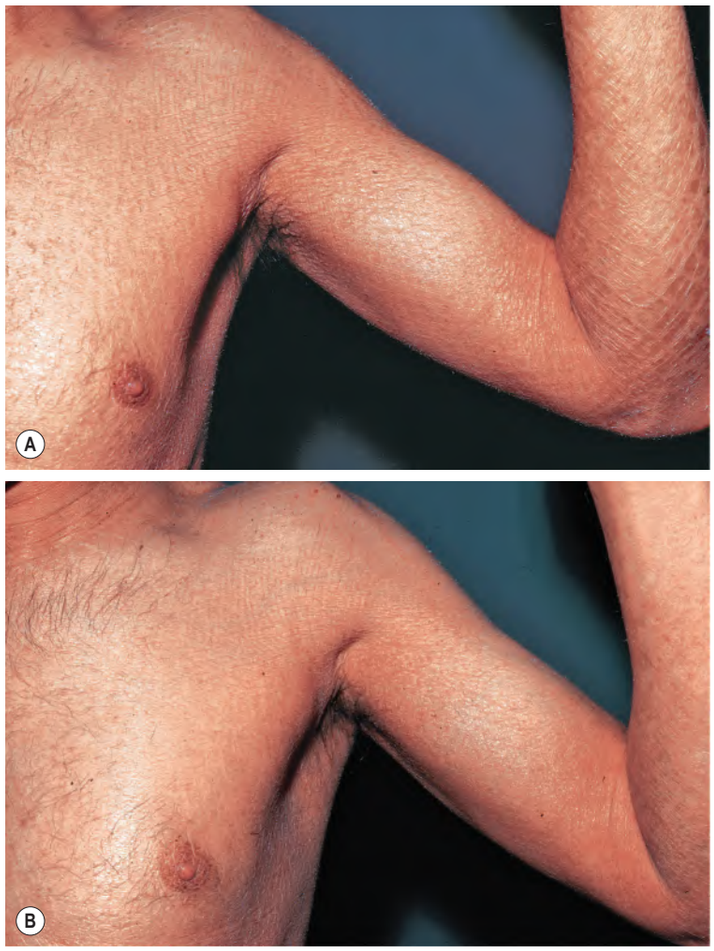

Fig. 126.5 Lamellar ichthyosis. Before (**A**) and after (**B**) treatment with acitretin.

isotretinoin 20 mg/day) in order to prevent an initial exacerbation of the disease; 20 mg/day of acitretin is usually sufficient for significant improvement.

Pityriasis rubra pilaris

Treatment with an oral retinoid (acitretin, isotretinoin, or alitretinoin) can lead to clearance of pityriasis rubra pilaris (PRP) (Fig. 126.6; see Ch. 9). In extensive cases, concomitant use of methotrexate may be advantageous, but with attention to possible liver toxicity.[53]

Rosacea

In severe or treatment-resistant rosacea, isotretinoin typically has a greater effect on inflammatory lesions than on vascular manifestations[54]. A low daily dose (10 mg) is often sufficient, although higher-dose acne vulgaris regimens are sometimes used for severe disease.

Premalignant and malignant skin lesions

Acitretin has been shown to be effective in the treatment of premalignant skin lesions, including human papillomavirus-induced neoplasia and actinic keratoses. In patients with basal cell nevus syndrome and xeroderma pigmentosum as well as solid organ transplant recipients, it may reduce the development of keratinocyte carcinomas (see Ch. 108). A prospective, open, randomized crossover study demonstrated that acitretin (25 mg/day for 12 months) prevented the development of squamous cell carcinoma in renal transplant recipients[55].

Lupus erythematosus

Both isotretinoin and acitretin have been used successfully in patients with various forms of lupus erythematosus (LE), including hypertrophic LE.[56] Recommended doses range from 0.2–1.0 mg/kg/day. Recurrence after completion of treatment is a limiting factor. Comparable therapeutic efficacy has been observed for acitretin and hydroxychloroquine in chronic and subacute cutaneous LE[57].

DOSAGE

The different formulations, strengths, and standard dosage ranges of topical and systemic retinoids currently available for use in dermatology are outlined in Table 126.5.

CONTRAINDICATIONS

Topical Retinoids

Topical retinoids should be avoided during pregnancy and nursing, due primarily to medicolegal issues raised by the teratogenicity of

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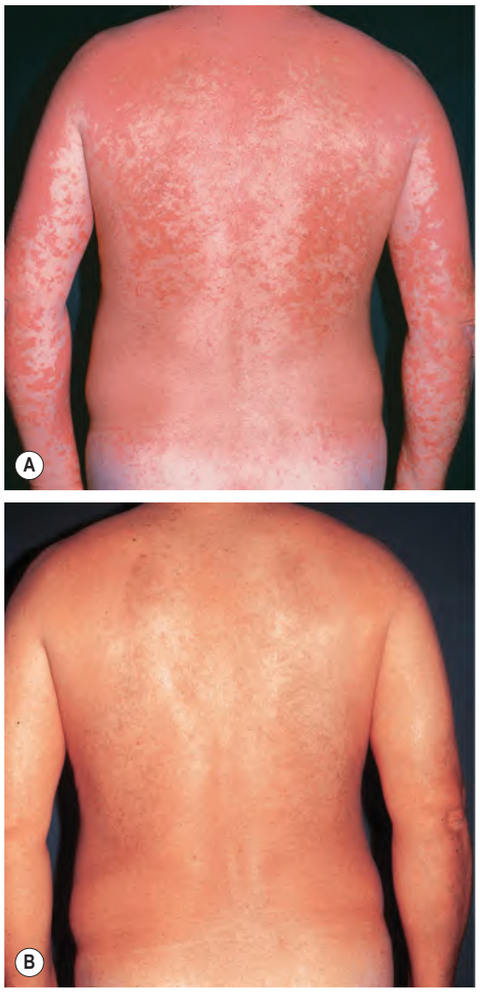

Fig. 126.6 Pityriasis rubra pilaris. Before (**A**) and after (**B**) treatment with acitretin.

oral retinoids. Use of irritating topical products (e.g. abrasive cleansers, astringents) and hair removal via waxing should be avoided in areas being treated with a topical retinoid.

Systemic Retinoids

The absolute contraindications are pregnancy, women contemplating becoming pregnant or non-compliant with contraception, breastfeeding, and hypersensitivity to medication components. Relative contraindications include moderate to severe hypercholesterolemia or hypertriglyceridemia, significant hepatic dysfunction (especially for bexarotene), hypothyroidism (especially for bexarotene and alitretinoin), suicidal ideation, and pseudotumor cerebri. Patients should be advised not to take vitamin A supplements in excess.

MAJOR SIDE EFFECTS

Topical Retinoids

By far the most common side effect of topical retinoids is skin irritation characterized by erythema and peeling (Table 126.6). "Retinoid dermatitis" typically occurs during the first month of treatment and tends

ADVERSE EFFECTS OF TOPICAL RETINOIDS	
Skin irritation (retinoid dermatitis)	Erythema Peeling, scaling Dryness Burning Pruritus Photosensitivity
Uncommon	Hypo- or hyperpigmentation Ectropion Allergic contact dermatitis Sticky skin

Table 126.6 Adverse effects of topical retinoids.

FORMULATIONS OF TOPICAL AND SYSTEMIC RETINOIDS				
Retinoid	**Route**	**Formulations**	**Standard dosage range**	**Major indications**
Retinol	Topical	Various (0.15%, 0.3%)		Cosmeceutical products
Retinaldehyde	Topical	Various (0.05%–0.1%)		
Tretinoin†	Topical	Gel, cream, solution (0.1%, 0.08%, 0.06%, 0.05%, 0.04%, 0.025%, 0.02%, 0.01%)		Acne vulgaris Photoaging
	Oral	10 mg capsules	45 mg/m^2/day	Acute promyelocytic leukemia
Tazarotene‡	Topical	Gel, cream, lotion, foam (0.045%, 0.05%, 0.1%)		Acne vulgaris Plaque-type psoriasis Photoaging
Adapalene¶	Topical	Gel, cream, lotion (0.1%, 0.3%)		Acne vulgaris
Trifarotene	Topical	Cream (0.005%)		Acne vulgaris
Isotretinoin	Topical§	Gel (0.05%)		Acne vulgaris*
	Oral	8, 10, 16, 20, 24, 25, 30, 35, 40 mg capsules	0.5–2 mg/kg/day	Severe, recalcitrant acne; related dermatoses*
Acitretin	Oral	10, 17.5, 22.5, 25 mg capsules	25–50 mg/day	Psoriasis Disorders of cornification*
Bexarotene	Topical	Gel (1%)		Cutaneous T cell lymphoma
	Oral	75 mg capsules	300 mg/m^2/day	
Alitretinoin	Topical	Gel (0.1%)		Kaposi sarcoma
	Oral§	10, 30 mg capsules	10–30 mg/day	Chronic hand eczema*

*Not an FDA-approved indication.
†Tretinoin is also available in combination with clindamycin phosphate 1.2% for acne or with fluocinolone acetonide 0.01% plus hydroquinone 4% for melasma.
‡Tazarotene 0.045% is also available in combination with halobetasol propionate 0.01% for plaque psoriasis.
¶Adapalene 0.1% is available over the counter and adapalene 0.1% and 0.3% are also available in combination with benzoyl peroxide 2.5% for acne.
§Currently not available in the US.

Table 126.5 Formulations of topical and systemic retinoids. Systemic retinoids are teratogenic and therefore contraindicated during pregnancy. Although topical retinoids have very low levels of systemic absorption, their use during pregnancy is not recommended and per the FDA, topical tazarotene and bexarotene are contraindicated during pregnancy. Topical tretinoin may be oxidized and inactivated by benzoyl peroxide, so it is recommended that these medications be applied at different times.

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to recede thereafter. It usually responds to a temporary reduction in the frequency, amount, or duration (e.g. "short contact" therapy) of retinoid application and to the use of moisturizers. Desquamation and peeling correspond to the hyperproliferative response of the epidermis to tretinoin mediated by RARs, but the erythema does not seem to be receptor-mediated. The perioral area of the face is most sensitive to peeling, and retinoid use in this region can be limited or avoided.

Although no photoallergic or phototoxic reactions have been proven for topical retinoids, some patients note a decreased tolerance to UVR. Erythema tends to develop shortly after sun exposure and is often accompanied by a sensation of heat, raising the question of involvement of infrared irradiation. The potential for teratogenicity from use of topical retinoids is very low to non-existent. Systemic absorption of topically applied retinoids has been inconsequential in both animal and human studies[58–60], and there is no evidence that topical application of tretinoin during pregnancy causes congenital disorders. Temporary worsening of acne may occur within the first weeks of therapy. Uncommon side effects include transient hypo- or hyperpigmentation, koebnerization of psoriasis (especially with tazarotene), allergic contact dermatitis, and rarely ectropion. Of note, topical retinoids have been used to treat ectropion in patients with ichthyosis.

Systemic Retinoids

Teratogenicity is the most concerning adverse effect of oral retinoids (Table 126.7). The side-effect profile of systemic retinoids (Table 126.8) resembles the toxic effects of vitamin A or hypervitaminosis A syndrome. Acute retinoid toxicity can include mucocutaneous (most common) and laboratory abnormalities, while chronic retinoid toxicity may occasionally produce bony changes.

Teratogenicity

Systemic retinoids are potent teratogens, and fetal deformities are the major concern in treating fertile women with oral retinoids. So far, with regard to teratogenicity, no safe minimal dose during pregnancy has been established. The defects seen in retinoid embryopathy can include craniofacial (microtia/anotia, micrognathia), cardiovascular, thymic, and central nervous system (CNS; hydrocephalus, microcephaly) abnormalities (see Table 126.7)[58,61]. These malformations may lead to spontaneous abortion, premature birth, or fetal death. The putative mechanism involves toxic effects on neural crest cells, particularly with exposure during the fourth week of gestation. No typical retinoid embryopathy malformations have been reported in pregnancies where only the male partner was taking acitretin or isotretinoin at the time of conception. Although it is often recommended that men who are actively trying to father children avoid systemic retinoid therapy due to

RETINOID EMBRYOPATHY
Craniofacial
• Atresia of the ear canal, microtia, anotia • Microphthalmia • Asymmetry of the face • Cleft palate • Micrognathia
Cardiovascular (especially conotruncal and aortic arch defects)
• Transposition of the great vessels • Tetralogy of Fallot • Truncus arteriosus • Supracristal ventricular septal defect • Interrupted aortic arch • Retroesophageal right subclavian artery • Hypoplastic aorta
Central nervous system
• Hydrocephalus • Microcephaly • Cortical agenesis • Cerebellar hypoplasia • Mental retardation • Difficulty with visual motor integration
Third and fourth pharyngeal pouches
• Thymic hypoplasia • Parathyroid hypoplasia
Other
• Limb anomalies • Anal and vaginal atresia

Table 126.7 Retinoid embryopathy.

ADVERSE EFFECTS OF SYSTEMIC RETINOIDS	
Acute effects	
Mucocutaneous	Xerosis with pruritus* Cheilitis* Dry mucosa*: mouth, nose (epistaxis), eyes (see below) Skin fragility* Retinoid dermatitis (see Table 126.6)* Palmoplantar peeling* Photosensitivity Sticky sensation (palms, soles) Granulation tissue with pyogenic granuloma-like lesions (e.g. at sites of acne cysts, periungual) Intertriginous erosions with exuberant granulation tissue Nail fragility with softening, onycholysis, paronychia Facial swelling *Staphylococcus aureus* infection Telogen effluvium, hair thinning
Ocular	Xerophthalmia* Blepharoconjunctivitis Reduced night vision Blurred vision, photophobia, keratitis, corneal ulceration (rare)
Systemic	Myalgias Arthralgias Anorexia, nausea, diarrhea, abdominal pain Headache, pseudotumor cerebri (rare) Fatigue, lethargy, irritability Depression, suicidal ideation (rare and controversial) Toxic hepatitis (rare) Clinical signs of hypothyroidism (with bexarotene*) Pancreatitis secondary to hypertriglyceridemia (rare) Hyperuricemia with gout (rare)
Laboratory	Elevated liver function tests (usually transient, minor): AST, ALT, alkaline phosphatase, LDH, bilirubin Hyperlipidemia: hypertriglyceridemia*; increased cholesterol, VLDL and LDL; decreased HDL Decreased free thyroxine (T4) (with bexarotene*) Elevated creatine kinase Leukopenia (with bexarotene) Agranulocytosis (with bexarotene) Thrombocytosis, thrombocytopenia Hypercalcemia (rare)
Chronic effects	
Mucocutaneous	Alopecia (rare) Dry eyes, corneal opacities (rare)
Systemic	DISH syndrome-like bone changes Osteophyte and bony bridge formation, especially of vertebrae (rarely clinically significant at dosages for acne) Anterior > posterior spinal ligament calcification Extraspinal tendon and ligament calcification Osteoporotic changes in long bones (rare, with possible exception of long-term etretinate) Premature epiphyseal closure (rare) Periosteal thickening Myopathy (rare)
*Common adverse effects.	

Table 126.8 Adverse effects of systemic retinoids. See Table 126.7 for teratogenicity. ALT, alanine aminotransferase; AST, aspartate aminotransferase; DISH, diffuse idiopathic skeletal hyperostosis; HDL, high-density lipoprotein; LDH, lactate dehydrogenase; LDL, low-density lipoprotein; VLDL, very-low-density lipoprotein.

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the presence of the drug in semen, only ~0.0001%–0.0005% of a typical oral dose would be delivered to a female partner (https://dailymed.nlm.nih.gov/).

In the US, physicians, patients (female or, to prevent sharing of medication, male), and pharmacies must register with a pregnancy risk reduction program (iPLEDGE™) in order for isotretinoin to be prescribed. The program requires female patients with childbearing potential to review an educational booklet, receive monthly counseling, take monthly comprehension and pregnancy tests (as well as two pregnancy tests prior to instituting therapy), and pledge to use two forms of effective contraception (or abstinence) continuously for 1 month before, during, and for 1 month after isotretinoin treatment[62,63]. Patients taking systemic retinoids should not donate blood while on therapy and for 1 month or 3 years after the last dose of isotretinoin or acitretin, respectively.

Skin and mucous membrane adverse effects

Dose-dependent mucocutaneous toxicity is the most commonly observed side effect of oral retinoids, and it reflects decreased production of sebum, reduced stratum corneum thickness, and altered skin barrier function (Fig. 126.7). Dry lips or cheilitis is the earliest and the most frequent finding (Fig. 126.8). Dryness of the mouth, nasal mucosa (associated with fragility and epistaxis), and eyes (see below) are other potential manifestations.

Xerosis of the skin and associated pruritus, peeling (especially of the palms and soles), and fissuring (particularly of the fingertips) are frequent side effects. Exacerbations of atopic dermatitis may occur. Photosensitivity is occasionally observed, in particular with isotretinoin, and probably reflects a reduction in the thickness of the stratum corneum. *Staphylococcus aureus* colonization correlates with isotretinoin-induced reduction in sebum production and may lead to cutaneous infections. Diffuse hair loss due to telogen effluvium is a relatively common complaint, although objective alopecia tends to occur only at high dosage levels. Effects on the nail apparatus can include thinning, fragility, and shedding of the nail plate as well as paronychia-like changes with periungual granulation tissue that can resemble pyogenic granulomas.

Fig. 126.7 Retinoid dermatitis. Xerosis, peeling, and erythema of the face in a patient treated with oral isotretinoin.

Fig. 126.8 Cheilitis in an isotretinoin-treated patient.

Mucocutaneous side-effect profiles vary among the systemic retinoids. Isotretinoin causes more mucosal dryness, and acitretin has been associated with higher incidences of alopecia and palmoplantar peeling, whereas bexarotene induces milder mucocutaneous and ocular side effects than other oral retinoids. Delayed wound healing and abnormal scarring during isotretinoin therapy have been described, and it has traditionally been suggested that elective procedures be postponed until 6–12 months after discontinuing isotretinoin when possible. However, recent systematic reviews and consensus recommendations do not support delaying most surgical and cosmetic procedures in patients currently receiving or who have recently completed isotretinoin therapy, with the exception of fully ablative laser treatments and mechanical dermabrasion[64]. Medium-depth and deep chemical peels are also often delayed.

Common laboratory abnormalities

Dyslipidemia

Hypertriglyceridemia is the most frequently observed systemic effect of retinoid therapy. Isotretinoin and etretinate/acitretin elevate triglycerides in 25%–50% and cholesterol in 20%–30% of treated patients[65], whereas bexarotene elevates triglycerides and cholesterol in ~80% and ~50% of patients, respectively[23]. A concomitant increase in total and low-density lipoprotein (LDL) cholesterol is frequently observed. In cases of severe retinoid-induced hypertriglyceridemia, eruptive xanthomas and acute hemorrhagic pancreatitis may occur. A higher incidence of acute hemorrhagic pancreatitis has been encountered in patients taking bexarotene, as compared to first- or second-generation retinoids.

Baseline fasting serum lipid levels should be obtained before initiating bexarotene therapy and every 1–2 weeks during therapy until levels become stable (generally in 4–8 weeks). For other oral retinoids, monitoring of lipid levels (together with liver function tests) is typically performed monthly for the first 2 months and then at 3-month intervals if baseline lipid levels are normal, there are no increases in dosage, and risk factors (obesity, high alcohol intake, diabetes) are absent. Less frequent monitoring (e.g. at baseline and after 2 months of treatment if results are normal) has been recommended for low-risk adolescents and young adults receiving isotretinoin for acne or ichthyosis, with subsequent testing every 6–12 months for ichthyosis patients on long-term treatment[52,66]. Discontinuation of therapy is suggested if fasting triglycerides reach 800 mg/dl (8 g/L). Less severe increases may be treated by dose reduction, withdrawing therapy until normalization of serum lipids occurs, and changes in diet or lifestyle. In some instances, lipid-lowering agents may be indicated. Coadministration of bexarotene with a statin (e.g. atorvastatin) for hypercholesterolemia and/or fenofibrate for hypertriglyceridemia is recommended to treat retinoid-induced hyperlipidemia and to lower the risk of pancreatitis[16,67]. Some authors recommend starting all patients on fenofibrate 7 days before initiating bexarotene treatment[37].

The effects of retinoid-induced hyperlipidemia during long-term therapy on the development of atherosclerotic cardiovascular disease is unknown. Retinoids are thought to cause hyperlipidemia by interfering with lipid clearance. Bexarotene increases the expression of apolipoprotein C-III, which prevents the uptake of lipids from very-low-density lipoproteins (VLDL) into cells.

Liver effects

Transient elevations in serum transaminases occur in ~20% of patients treated with acitretin and much less frequently in those receiving isotretinoin or bexarotene therapy. Circulating levels of alkaline phosphatase, lactic dehydrogenase, and bilirubin may also occasionally become elevated during retinoid therapy. Liver function test abnormalities are typically mild, usually developing between 2 and 8 weeks after starting therapy and then returning to normal within another 2–4 weeks despite continued therapy. Severe or persistent hepatotoxic reactions occur in <1% of patients. Acitretin therapy elicited no biopsy-proven hepatotoxicity in a 2-year prospective study, thus suggesting that periodic liver biopsy is not necessary[68]. No studies have specifically evaluated the use of retinoids in patients with hepatic insufficiency. However, since retinoids are metabolized by hepatic cytochrome P450 isoenzymes

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(e.g. CYP3A4) and undergo partial biliary elimination, significant hepatic insufficiency is likely to interfere with drug elimination. Transaminase elevations to greater than three times the upper limit of normal should prompt discontinuation of retinoid therapy. Other causes of hepatic dysfunction should be excluded.

Other systemic adverse effects

Ophthalmologic effects

The most common ocular retinoid effects are dryness and irritation. Xerophthalmia due to decreased Meibomian gland secretions may prohibit the use of contact lenses and can lead to blepharoconjunctivitis, keratitis, or corneal opacities (especially with isotretinoin). Alterations in visual function, mainly nyctalopia (reduced night vision), excessive glare sensitivity, and changes in color perception, have also been reported[69]. Competitive inhibition of ocular retinol dehydrogenase by retinoids, resulting in decreased rhodopsin formation, may be the cause of nyctalopia.

Bone effects

Bone pain can occur in retinoid-treated patients without objective evidence of any abnormalities and without sequelae. Several reports have also implicated therapy with oral retinoids, including etretinate and isotretinoin, in the formation of diffuse hyperostoses of the spine (diffuse idiopathic skeletal hyperostosis [DISH] syndrome-like changes) as well as calcification of tendons and ligaments, particularly in the ankles[70]. These changes are thought to be related to the dose and duration of retinoid treatment[52], with specific findings including anterior spinal ligament calcification, osteophyte ("spur") formation, extraspinal calcifications, and bony bridges, but not narrowing of the disk space. Of note, many of the bone changes observed are common in the general population (especially in adults) and not clinically significant. Prospective studies have shown that the likely potential effect of retinoids is to worsen pre-existing skeletal overgrowth rather than to induce *de novo* changes.

Osteoporosis has been observed with hypervitaminosis A and after long-term therapy with etretinate but not with acitretin or isotretinoin. Several investigations in patients receiving a standard isotretinoin course for acne (~1 mg/kg/day for 5–6 months), including an open-label trial in 217 adolescents and three controlled studies in adolescents (n = 21; n = 358) and young adults (n=36), showed no significant changes in bone mineral density[71,72]. Likewise, a study of 30 adults found no association between acitretin therapy (median duration, 3.6 years) for chronic dermatoses and risk of osteopenia or osteoporosis. Although there are rare reports of premature epiphyseal closure in children treated with oral retinoids (e.g. >1 mg/kg/day for ≥4–6 years)[52], this has not been observed in acne treatment studies. A baseline radiographic survey or bone density evaluation is not required prior to administering an oral retinoid, although monitoring high-risk patients who require prolonged high-dose therapy may be useful.

Muscle effects

Although muscular pain and cramps can occur in patients taking acitretin, these symptoms are more common with higher doses of isotretinoin therapy, especially in individuals involved in vigorous physical activity. Occasionally, elevated creatine kinase levels may be observed. Increased muscle tone as well as axial muscle rigidity and myopathy have been reported in patients receiving etretinate and acitretin therapy, respectively.

Central nervous system and psychiatric effects

Although individual signs of increased intracranial pressure, such as headache, nausea and vomiting, are occasionally observed in patients taking a systemic retinoid, the complete pseudotumor cerebri syndrome with papilledema and blurred vision is very rare[73]. Concomitant use of other drugs associated with intracranial hypertension (e.g. tetracyclines) is considered a risk factor for developing pseudotumor cerebri and should therefore be avoided[74]. Examination for papilledema should be performed immediately when a patient receiving retinoid therapy complains of a persistent headache, especially if it is accompanied by visual changes, nausea or vomiting, or when pseudotumor cerebri is otherwise suspected.

Although there have been anecdotal reports of depression, psychosis, and suicide attempts in acne patients taking isotretinoin, a causal link has not been established. Multiple prospective and population-based studies as well as case–control, prescription sequence symmetry, and meta-analyses have failed to find an increased risk of depression or suicidal behavior associated with isotretinoin therapy for acne[75–77]. In contrast, several studies and meta-analyses actually noted a significant *decrease* in depressive symptoms in individuals treated with isotretinoin compared to baseline or a control group receiving other acne therapies[75–77], while one small case-crossover study showed a positive correlation between recent exposure to isotretinoin and diagnosis of depression in patients with acne. Acne patients should be informed of possible associations between isotretinoin and psychiatric issues prior to initiating this treatment and monitored for depressive symptoms or suicidal ideation during therapy.

Hypothyroidism

Clinical and biochemical central hypothyroidism occurred in 40% of patients in CTCL trials with bexarotene and was rapidly and completely reversible with cessation of therapy[23,78]. Dose-dependent decreases in serum thyroid-stimulating hormone (TSH) levels were also noted in clinical trials of alitretinoin. This effect is probably mediated via suppressed secretion of the β-subunit of TSH by the thyrotrope cells of the anterior pituitary, which express RXR-γ. Free T4 levels should be monitored before and during bexarotene and alitretinoin therapy and normalized with thyroid hormone replacement as necessary[37].

Hematologic effects

A high incidence (~30%) of dose-related leukopenia was reported in studies of bexarotene for CTCL, occurring as early as 2–4 weeks and characterized by a decrease in neutrophils rather than lymphocytes[23]. Hematologic abnormalities are much less common with other retinoids, but monitoring the complete blood count in HIV-infected patients is warranted.

Gastrointestinal effects

Uncommon nonspecific gastrointestinal complaints have been reported in association with retinoid therapy. Although there have been case reports of isotretinoin administration preceding the onset of inflammatory bowel disease (IBD) and one study showed an association between previous isotretinoin use and the development of ulcerative colitis, a cause-and-effect relationship has not been established. Several large population-based studies found that patients with IBD were no more likely to have used isotretinoin prior to diagnosis than matched controls[79,80], and a meta-analysis concluded that isotretinoin use does not increase the risk of IBD overall or ulcerative colitis specifically[79]. Potential confounding effects of severe inflammatory acne itself and treatment with oral antibiotics may also contribute to an apparent association between isotretinoin and IBD.

Renal effects

Renal toxicity is not characteristic of retinoid therapy. Isotretinoin has been safely administered to patients with end-stage kidney disease who were undergoing hemodialysis. Renal function should be monitored in patients with a history of kidney disease.

Summary

Most adverse effects associated with retinoids are preventable or manageable through proper patient selection, diligent monitoring, and dose adjustments (or occasionally discontinuation of treatment) as indicated.

INTERACTIONS

Concurrent use of systemic retinoids with alcohol or other medications having similar side effects may increase the incidence of adverse events. The following should be avoided or used with caution:

- tetracycline, doxycycline, minocycline (may increase intracranial pressure)
- alcohol (increased conversion of acitretin to etretinate and hepatotoxicity)
- methotrexate (synergistic liver toxicity with retinoids; however, combination may be used with caution in patients with PRP or severe psoriasis)
- vitamin A supplements (risk of hypervitaminosis A).

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Retinoid drug levels and resultant potential for toxicity may increase with CYP3A4 inhibitors such as azoles and macrolides. In contrast, medications and herbal supplements including antituberculous drugs (e.g. rifampin), anticonvulsants (e.g. phenytoin, carbamazepine), and St. John's wort may decrease the drug levels of retinoids via CYP3A4 induction (see Table 131.8). Retinoids may also increase the drug levels of cyclosporine via competition for CYP3A4 metabolism[16]. Concomitant administration of bexarotene and gemfibrozil results in a substantial increase in plasma concentrations of bexarotene, at least partially because of cytochrome P450 CYP3A4 inhibition that is induced by gemfibrozil. Concomitant administration of fenofibrate (for hypertriglyceridemia) or atorvastatin (for hypercholesterolemia) does not affect bexarotene plasma levels and, therefore, can be utilized together with bexarotene (see Table 126.4).

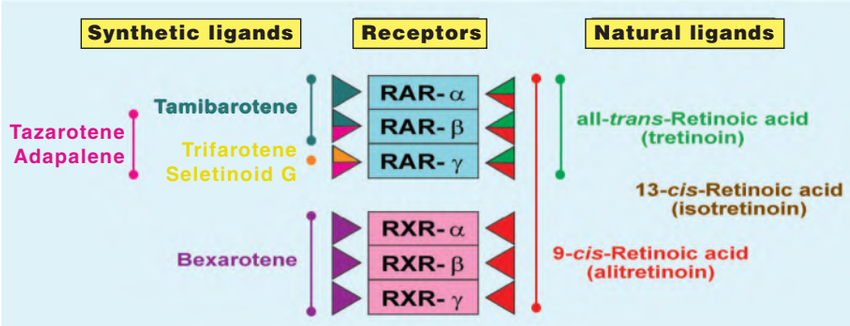
In rare instances, patients with diabetes mellitus may have more difficult glucose control while taking a retinoid; however, the causal relationship remains uncertain. Acitretin has been shown to possibly reduce the efficacy of progestin-only contraceptives. Also, unprotected UVR exposure or photosensitizing medications should be avoided because of the potential for increased photosensitivity related to retinoid therapy. Since retinoids act through nuclear receptor dimers, cross-talk with other nuclear receptors may provide synergistic effects: for example, interactions between RXR and vitamin D receptor (VDR) positively influence the action of vitamin D_3, and cross-talk with PPAR and its ligands is under investigation.

USE IN PREGNANCY AND LACTATION

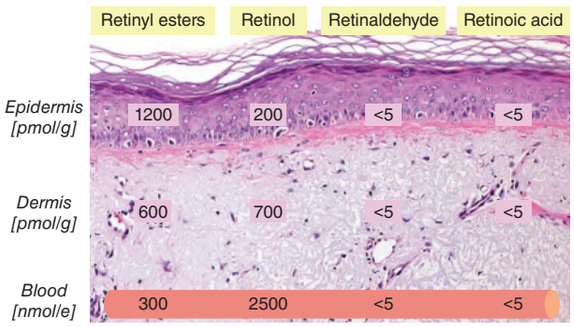
All systemic retinoids are teratogenic and are absolutely contraindicated during pregnancy and lactation (formerly FDA category X). As discussed above, pregnancy testing is necessary for female patients of childbearing potential before instituting oral retinoid therapy (two tests required in the US for isotretinoin) and then at regular intervals (e.g. monthly for isotretinoin) during its administration. Adequate contraception for at least 1 month before initiation and throughout the treatment course is also mandatory in this patient group. Acitretin requires contraception for 3 years (US) or 2 years (Europe) after cessation of therapy, whereas isotretinoin, bexarotene, and alitretinoin require contraception for just 1 month after cessation of therapy.

Although no evidence exists for teratogenicity of topical retinoids in humans, they are nevertheless not recommended for use during pregnancy. Since no dermatologic condition that is known to be responsive to topical retinoids is life-threatening to the mother or fetus, it is wiser to postpone treatment until after delivery, primarily for medicolegal reasons rather than scientific data in the opinion of the authors. As it is not known whether topical retinoids pass into the breast milk, topical retinoid use is generally not recommended during breastfeeding.

Additional figures available in our eBook (see inside front cover for access code).



eFig. 126.1 Binding properties of natural and synthetic retinoids on nuclear retinoid receptors.



eFig. 126.2 Mean physiologic concentrations of natural retinoids in the skin and blood. Vitamin A (retinol) is found in the skin mainly esterified to fatty acids, whereas unesterified retinol is the major circulating form.

eFig. 126.3 Osteophytes and bony ridges without narrowing of the disk spaces in an acne patient treated with a prolonged (several-year) course of isotretinoin.

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eFig. 126.4 Severe acne vulgaris. Before (**A**) and after (**B**) treatment with oral isotretinoin.

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